

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

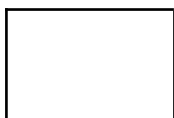
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

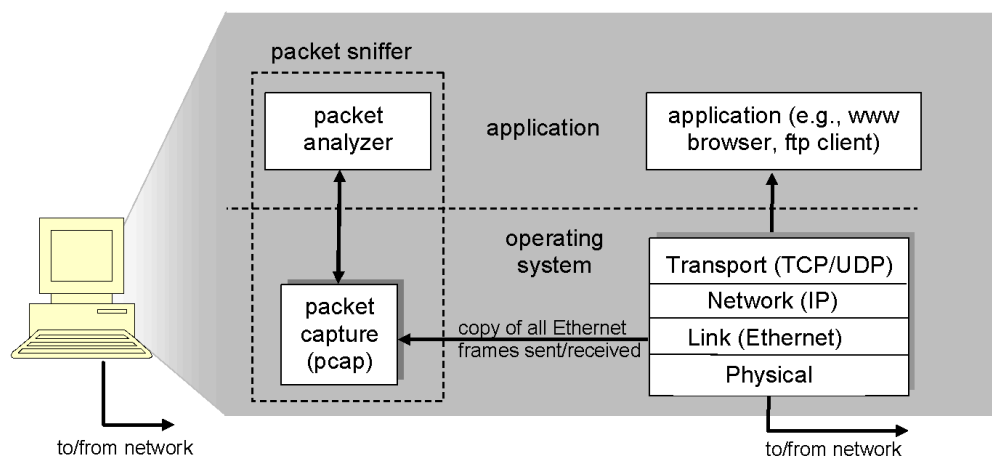


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

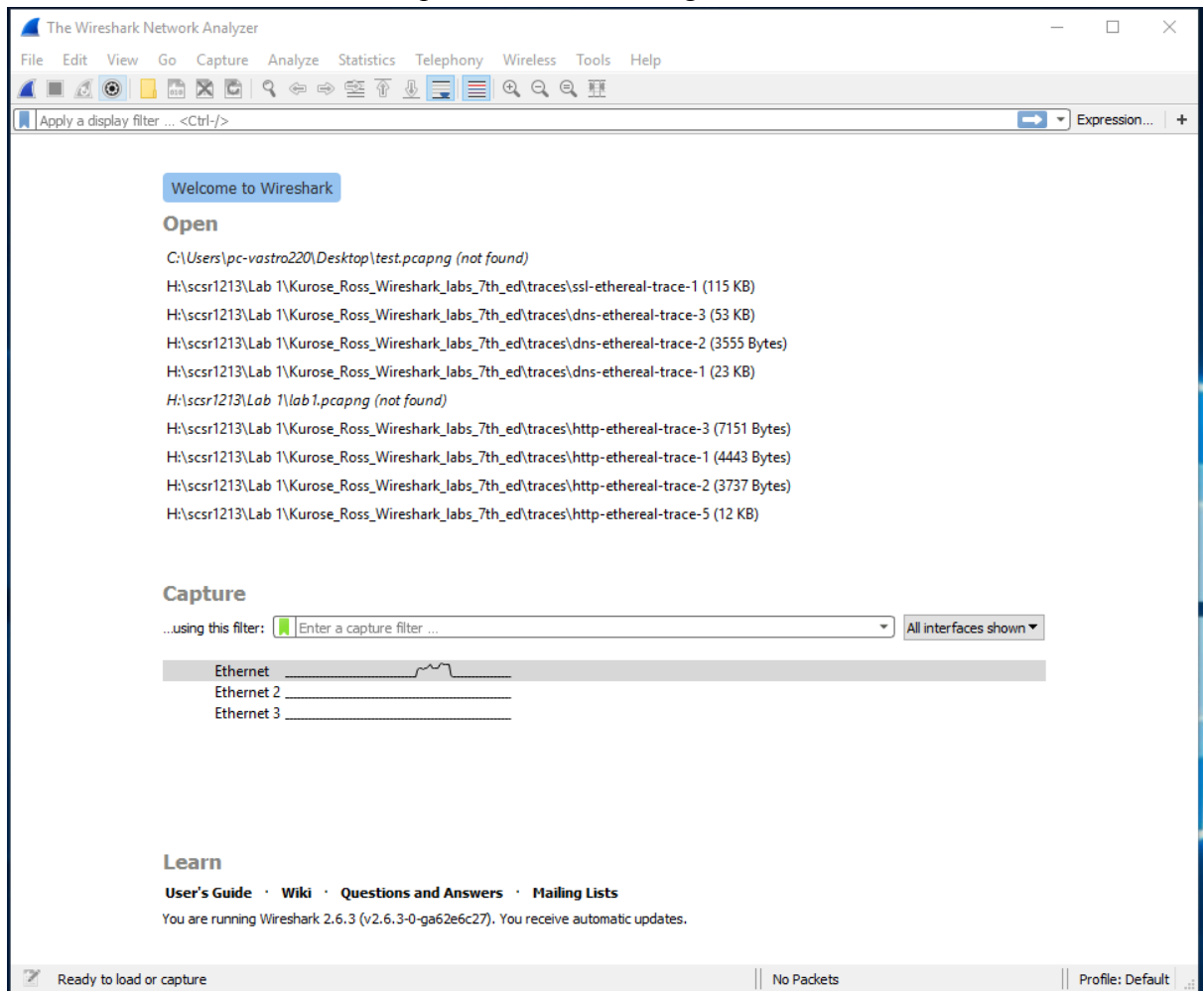


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

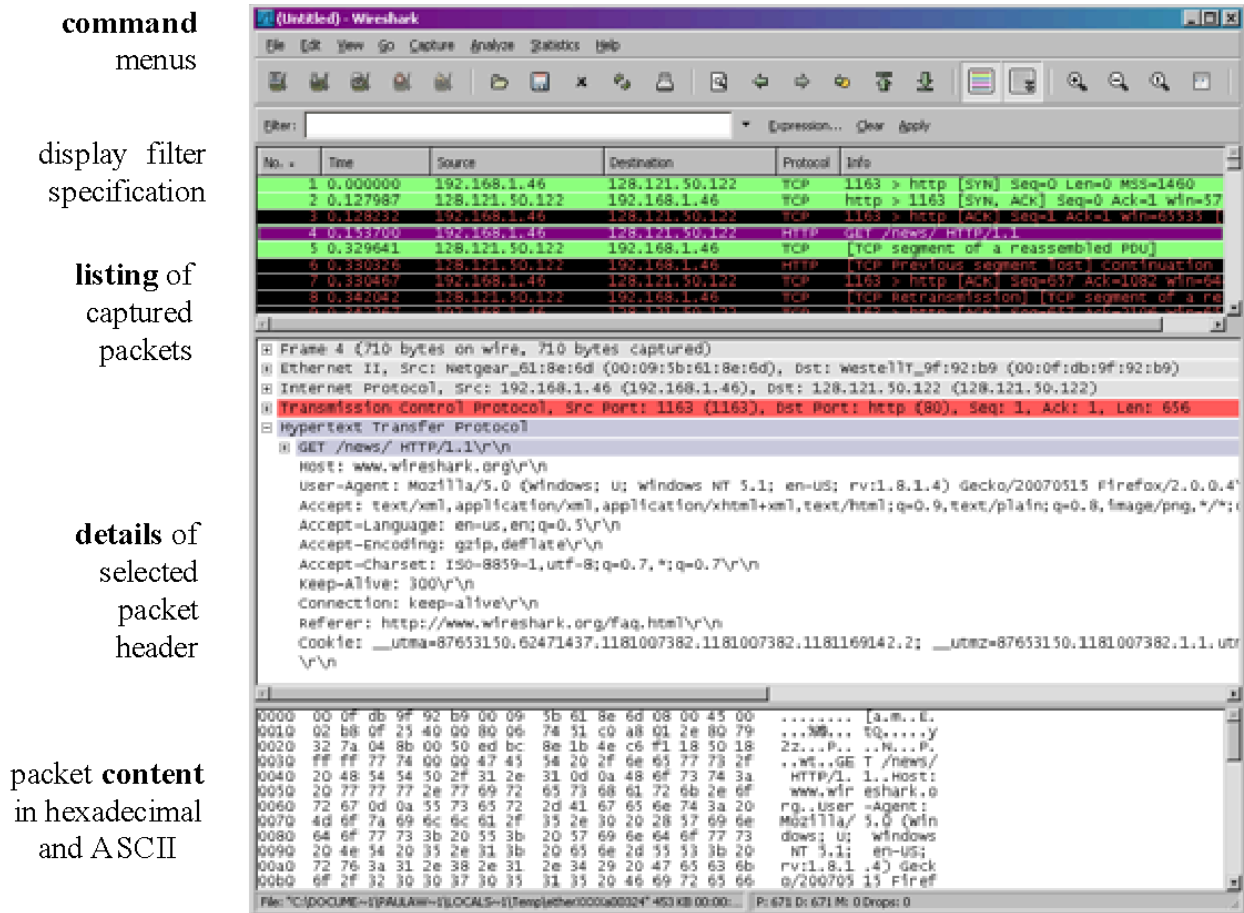


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

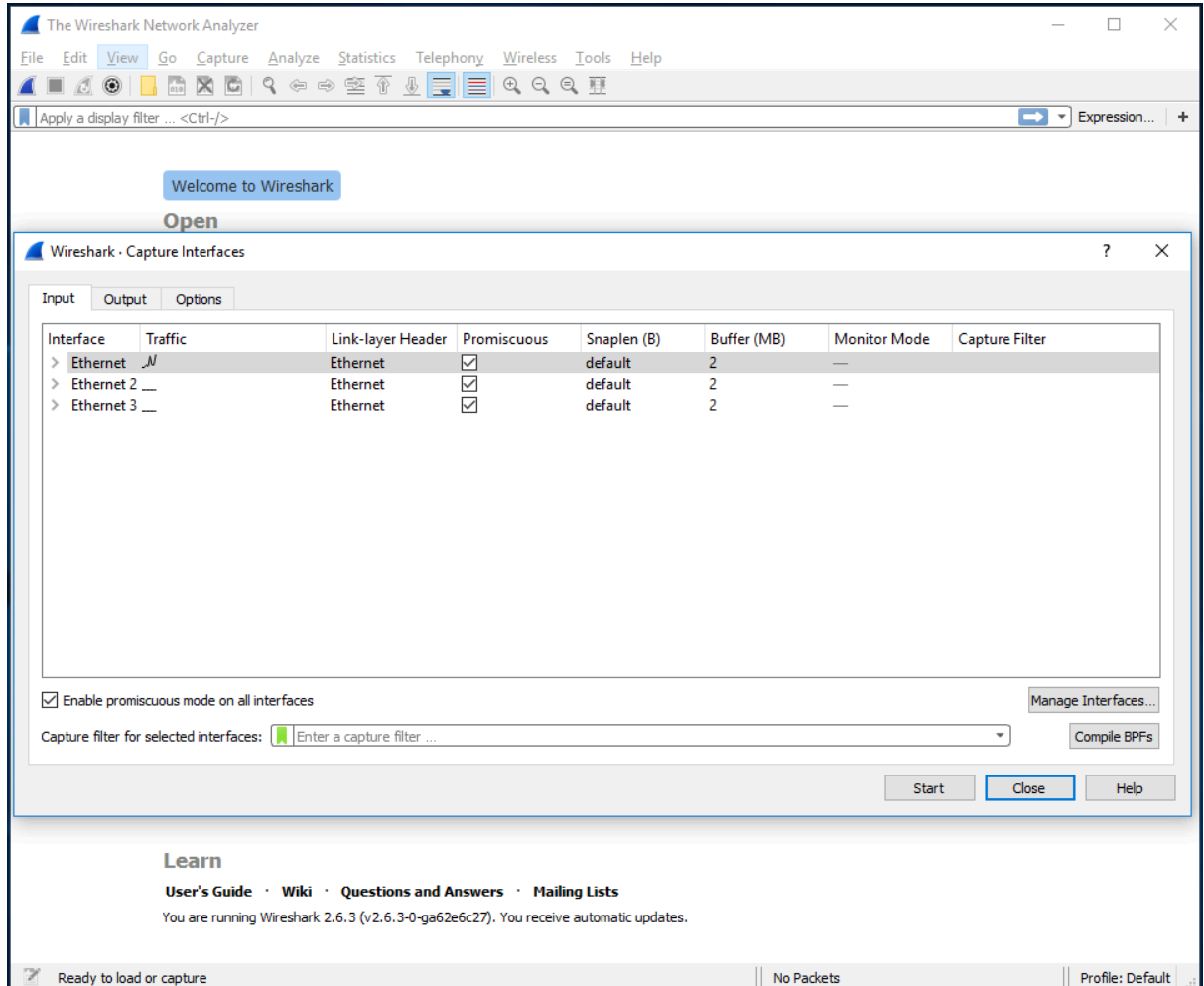


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

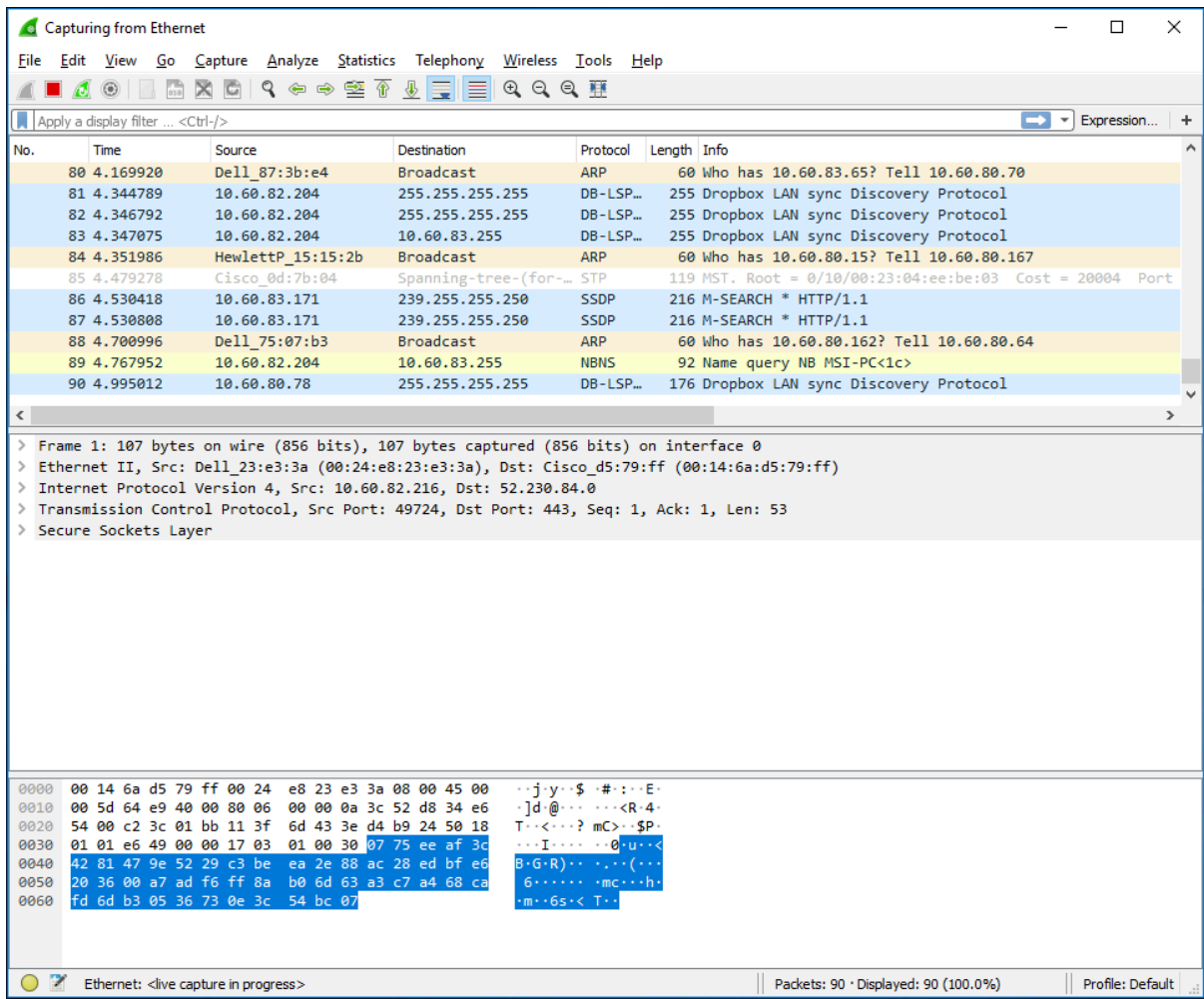


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

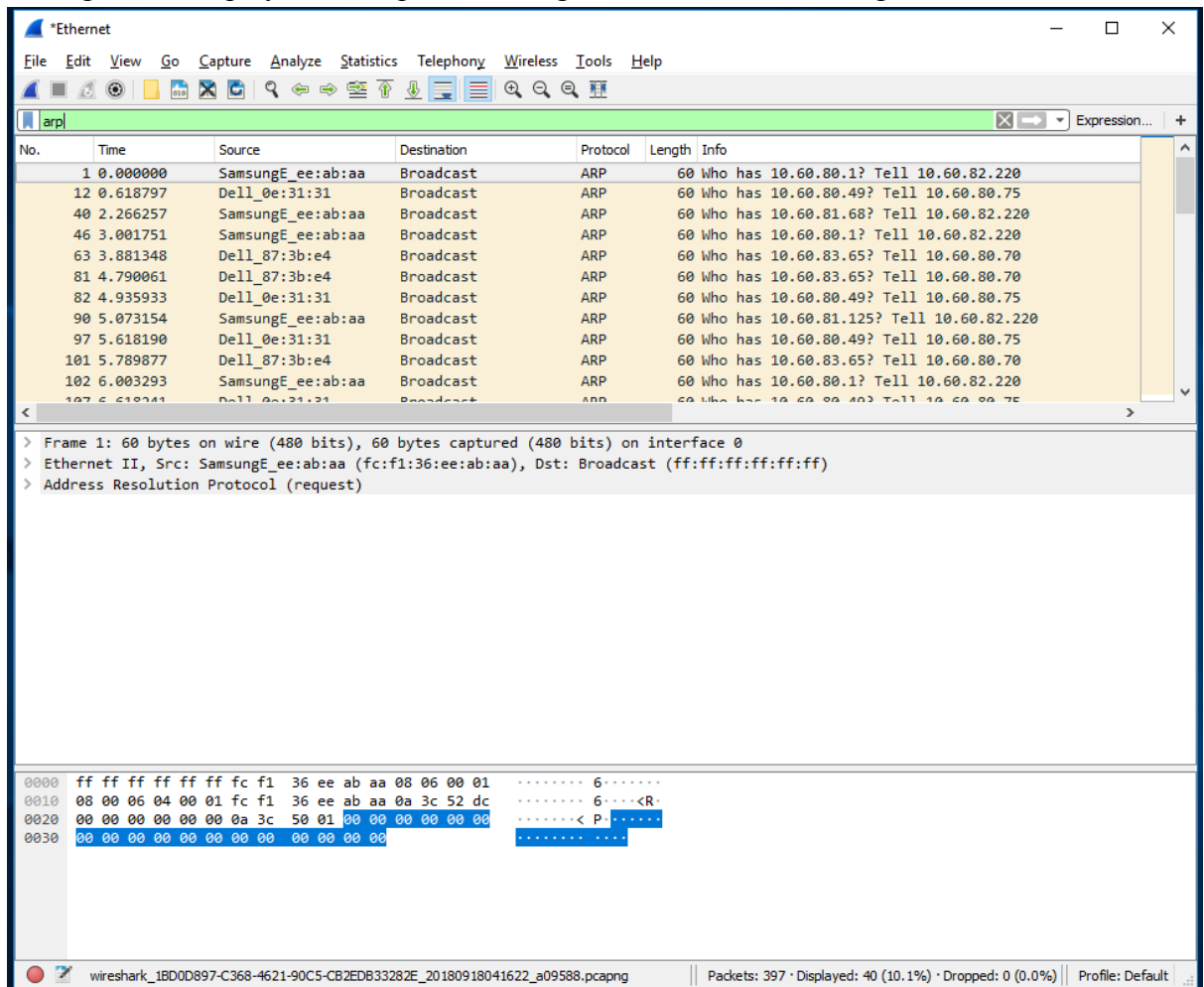


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

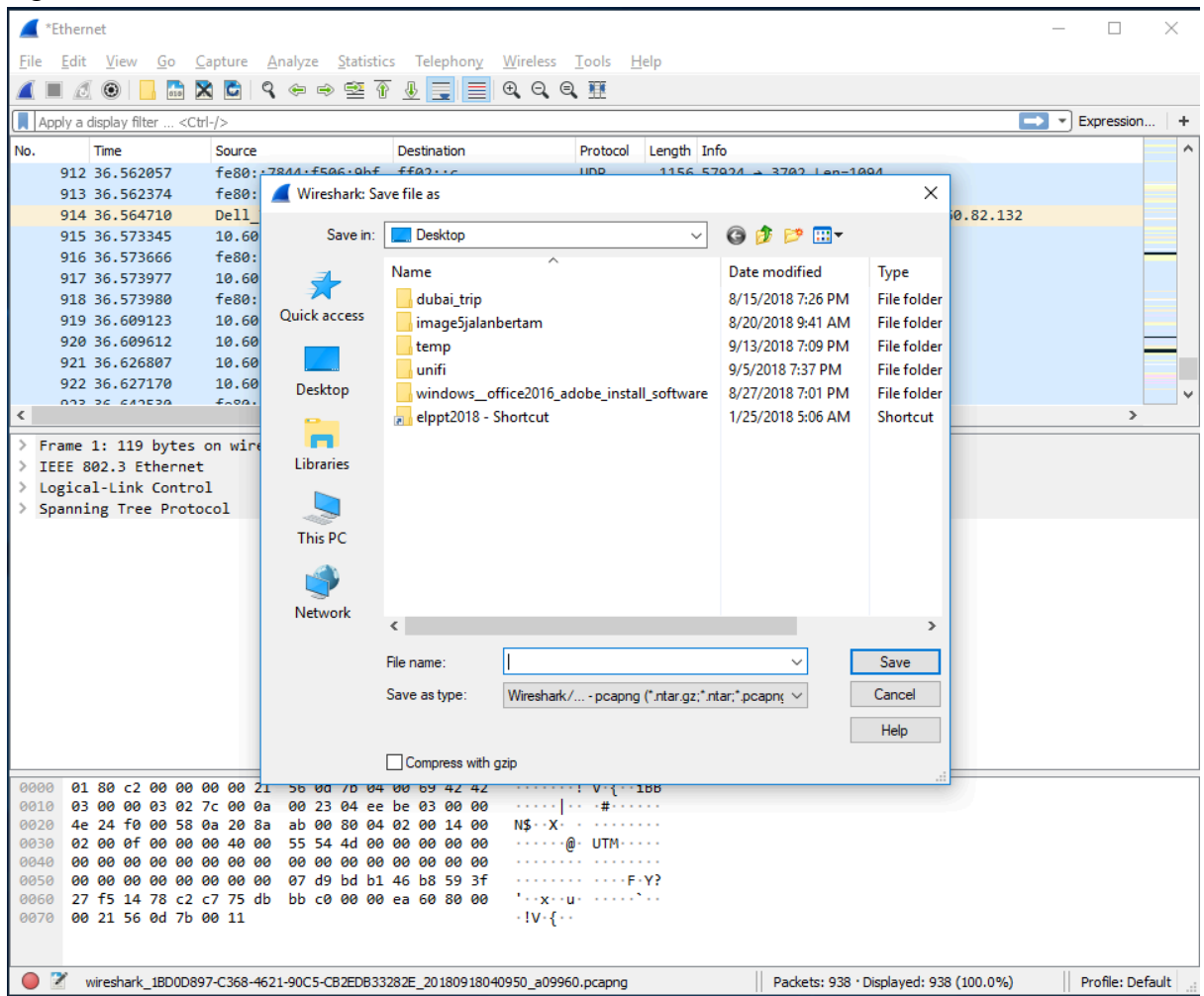


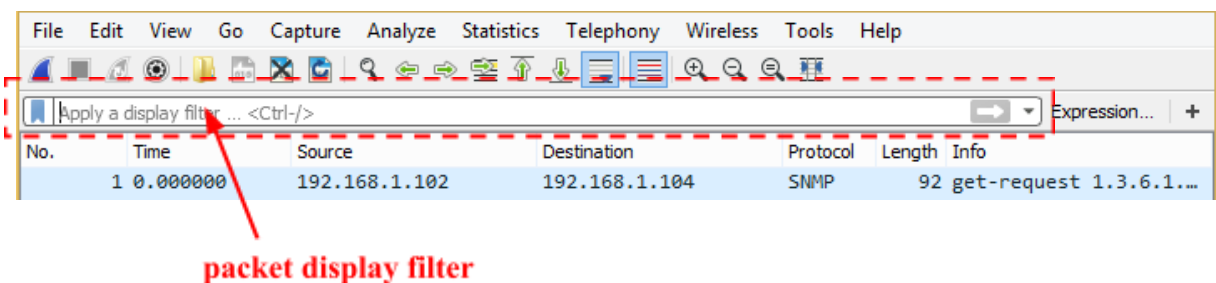
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?

HTTP/1.1

2. What is the IP address of the client computer?

192.168.1.102

3. What is the IP address of the gaia.cs.umass.edu server?

128.119.245.12

4. How many bytes of content are being returned to client browser?

73 bytes

5. What is the status code returned from the server to client browser?

200 OK for GET Request URI: /ethereal-labs/lab2-1.html and 404 Not Found for GET Request URI: /favicon.ico

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes. Because the status code is 200 OK

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes. If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The status code is 304 Not Modified. No, the server does not explicitly return the contents of the file because the requested file has not changed since the last time the client accessed it.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

Three times

2. To which Internet addresses were these GET requests sent?

128.119.245.12, 165.193.123.218 and 134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

3357 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

15642 bytes

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\Owner>nslookup www.microsoft.com
Server:  ns1.utm.my
Address: 161.139.250.2
```

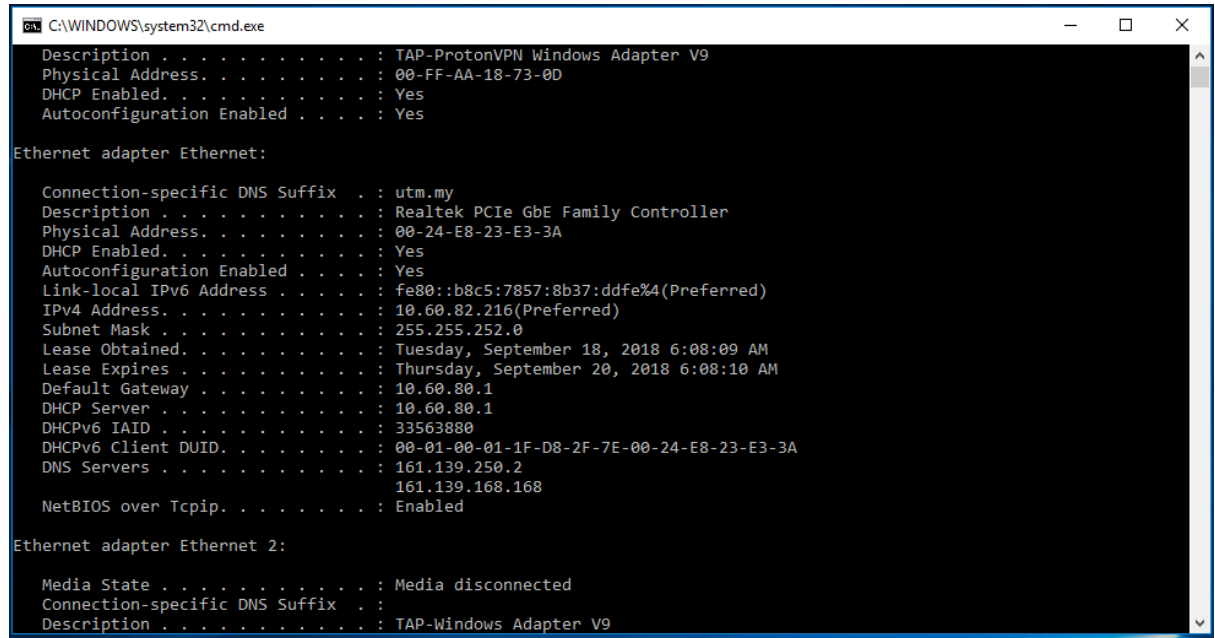
2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
Non-authoritative answer:
Name:      e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c84::356e
           2001:f40:a:c81::356e
           211.25.121.218
Aliases:   www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes

Ethernet adapter Ethernet:

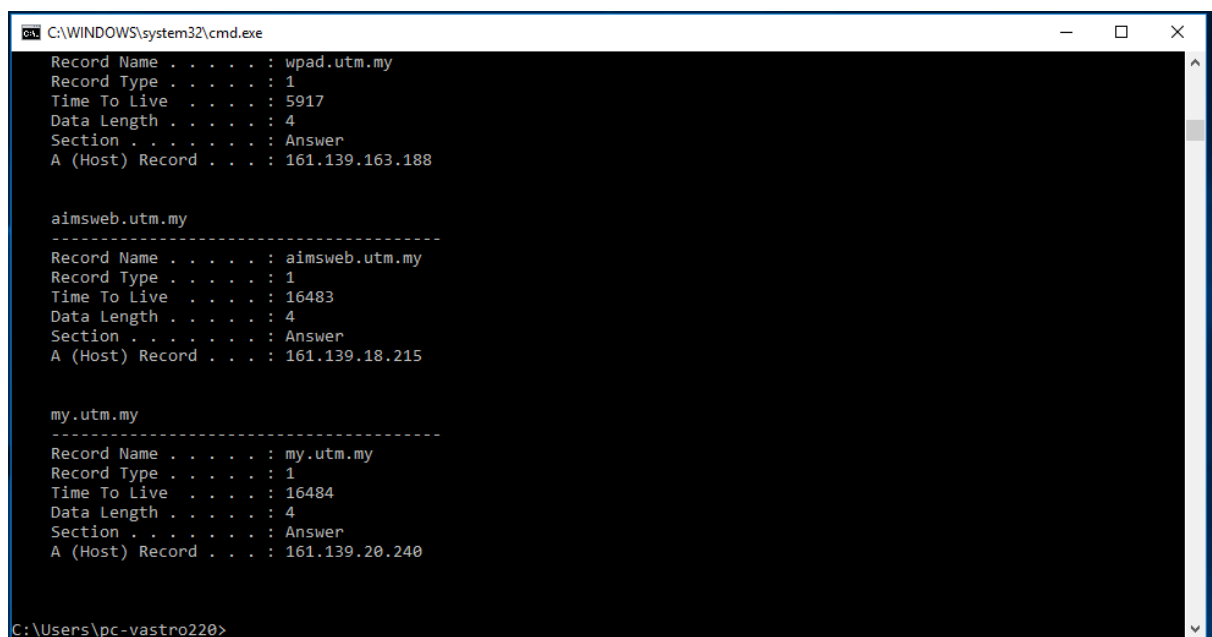
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

aimswb.utm.my
-----
Record Name . . . . . : aimswb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

my.utm.my
-----
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



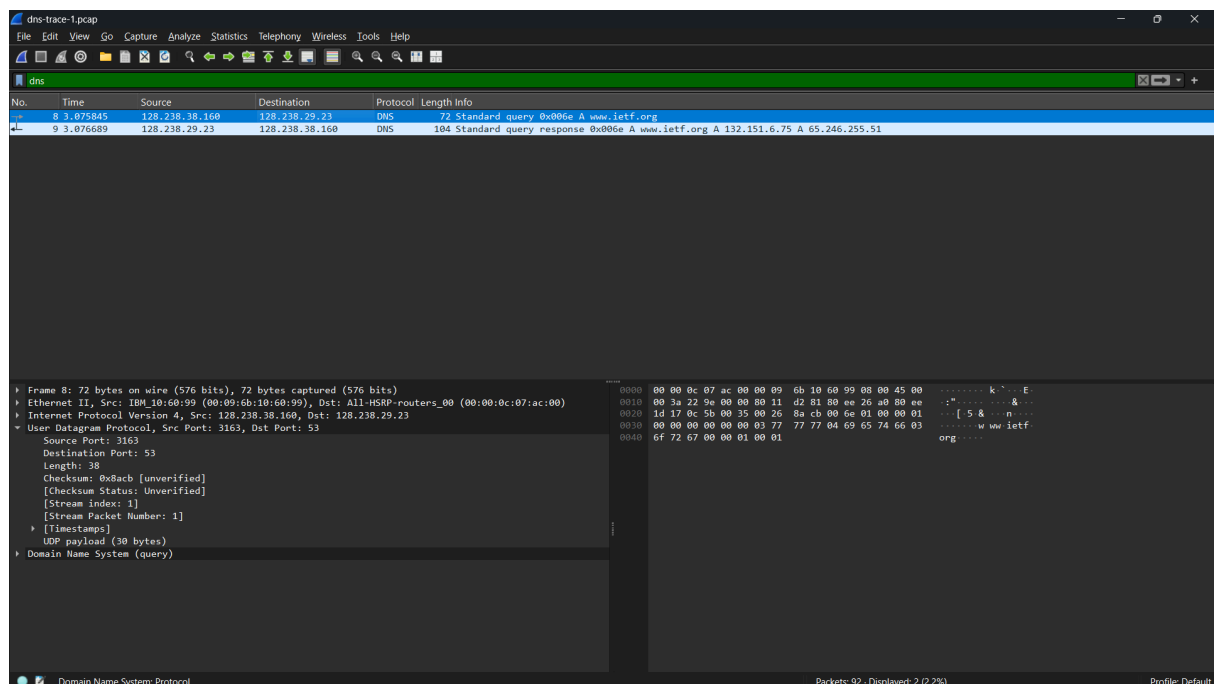
```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns
Windows IP Configuration
Successfully flushed the DNS Resolver Cache.
C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1.pcap. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

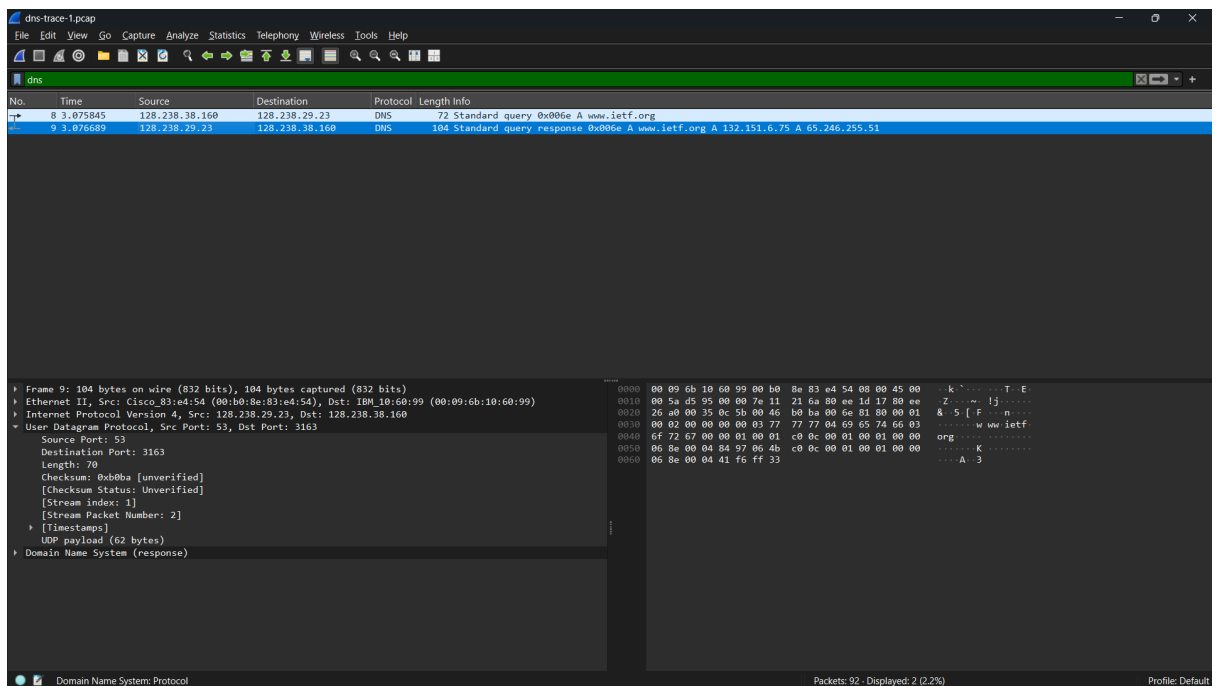
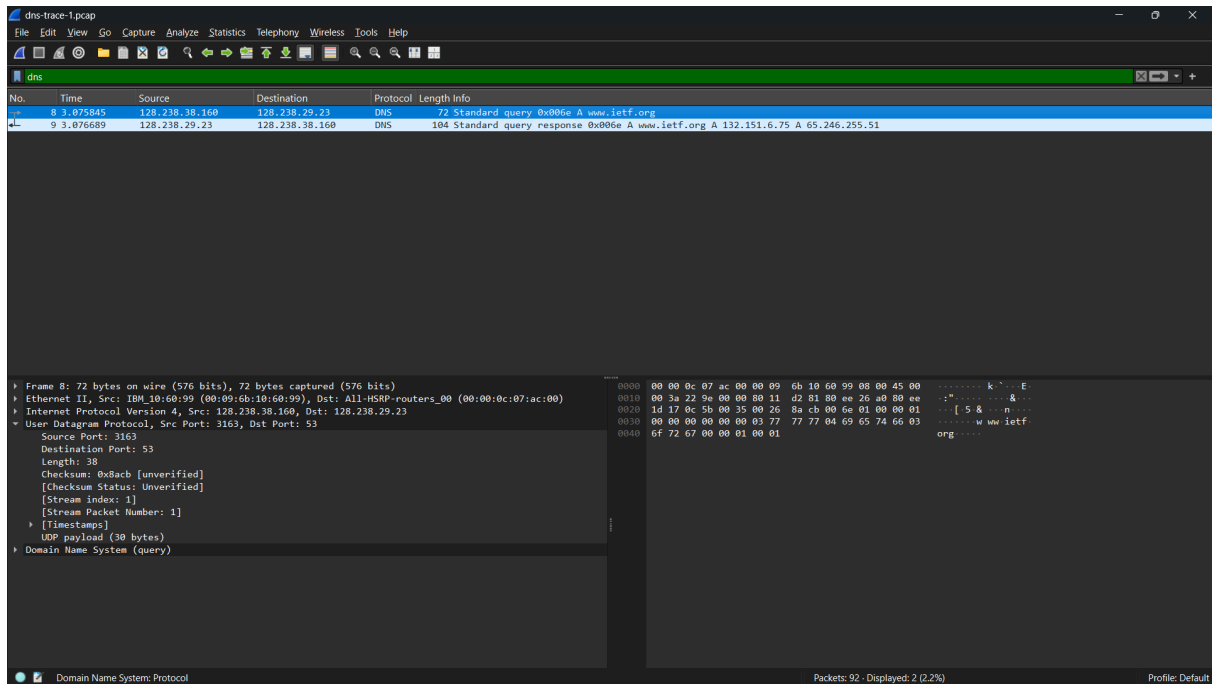
UDP



2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

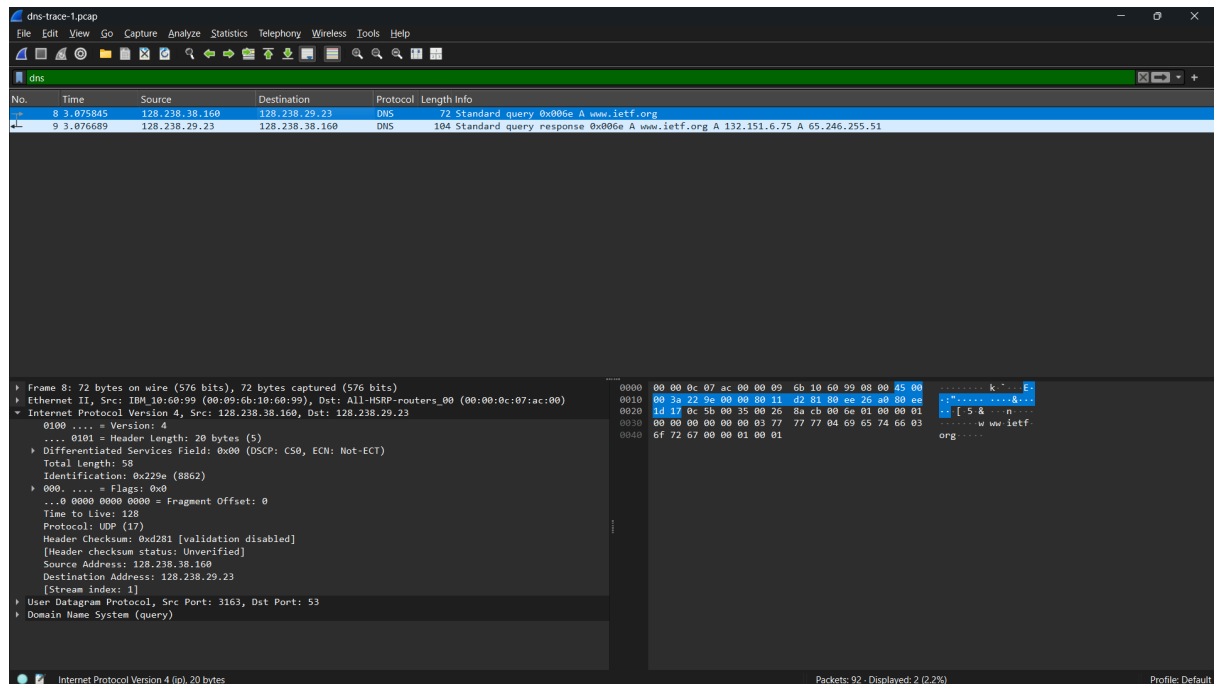
DNS query destination port: 53

DNS response source port: 53



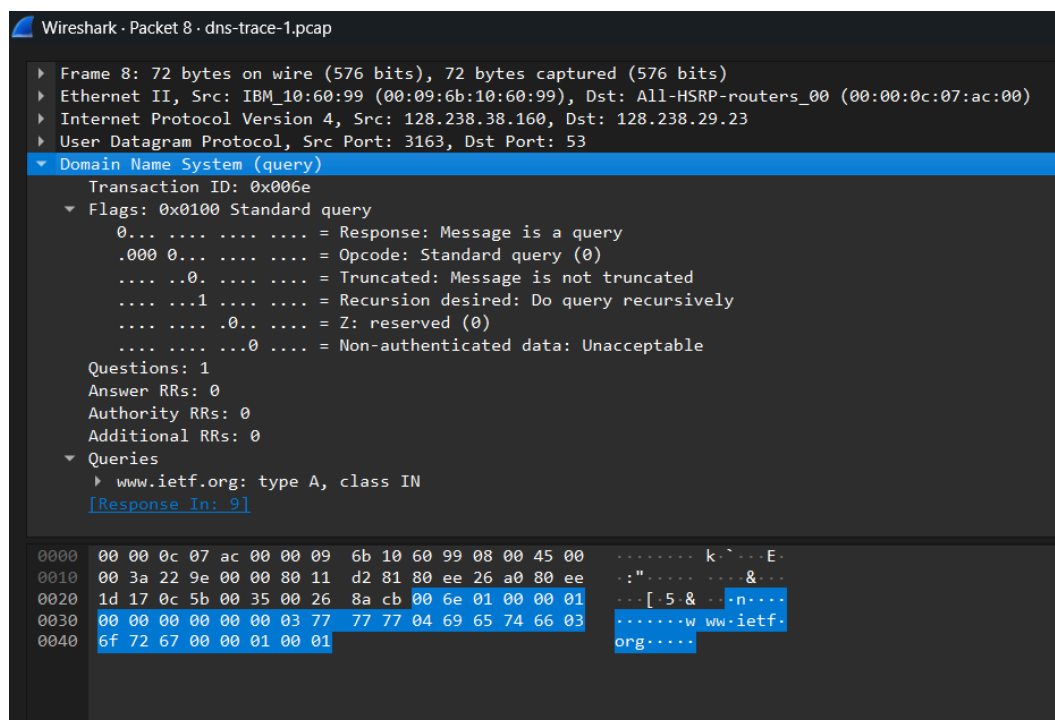
3. To what IP address is the DNS query message sent? Add screenshots in your answer.

128.238.29.23



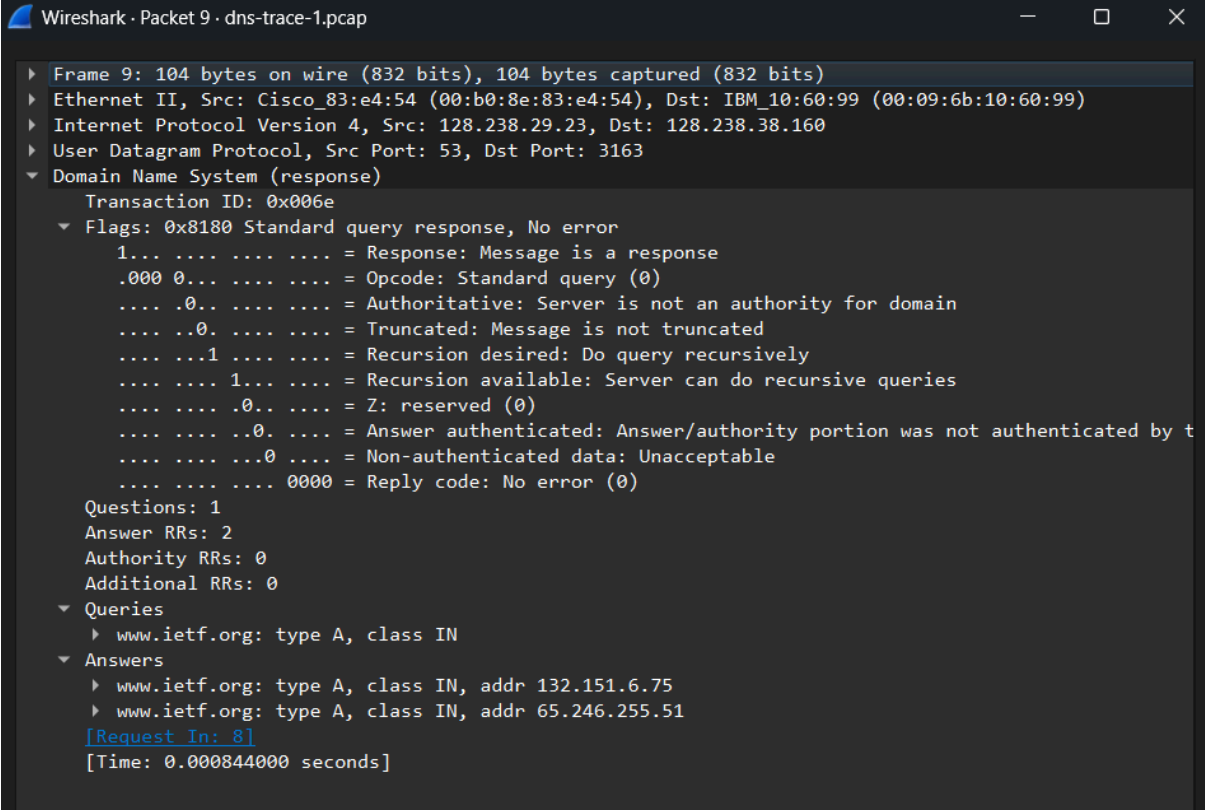
4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type A, No “answers”



5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

2 “answers”. Address 132.151.6.75 and 65.246.255.51

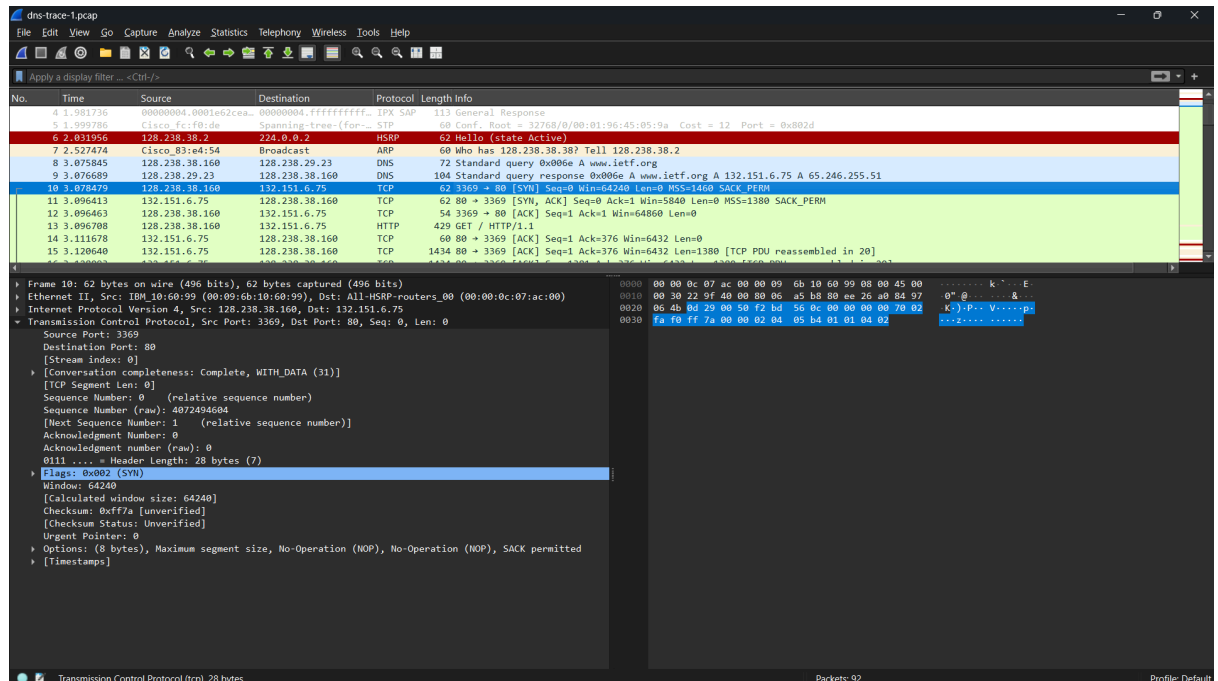


The screenshot shows a Wireshark packet capture window titled "Wireshark · Packet 9 · dns-trace-1.pcap". The packet list on the left shows "Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)". The packet details pane on the right shows the following structure:

- ▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
- ▶ Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
- ▶ User Datagram Protocol, Src Port: 53, Dst Port: 3163
- ▼ Domain Name System (response)
 - Transaction ID: 0x006e
 - ▼ Flags: 0x8180 Standard query response, No error
 - 1... .. = Response: Message is a response
 - .000 0... .. = Opcode: Standard query (0)
 -0... .. = Authoritative: Server is not an authority for domain
 -0... .. = Truncated: Message is not truncated
 -1... .. = Recursion desired: Do query recursively
 -1... .. = Recursion available: Server can do recursive queries
 -0... .. = Z: reserved (0)
 -0... .. = Answer authenticated: Answer/authority portion was not authenticated by t
 -0... .. = Non-authenticated data: Unacceptable
 -0000 = Reply code: No error (0)
 - Questions: 1
 - Answer RRs: 2
 - Authority RRs: 0
 - Additional RRs: 0
 - ▼ Queries
 - ▶ www.ietf.org: type A, class IN
 - ▼ Answers
 - ▶ www.ietf.org: type A, class IN, addr 132.151.6.75
 - ▶ www.ietf.org: type A, class IN, addr 65.246.255.51
 - [\[Request In: 8\]](#)
 - [Time: 0.000844000 seconds]

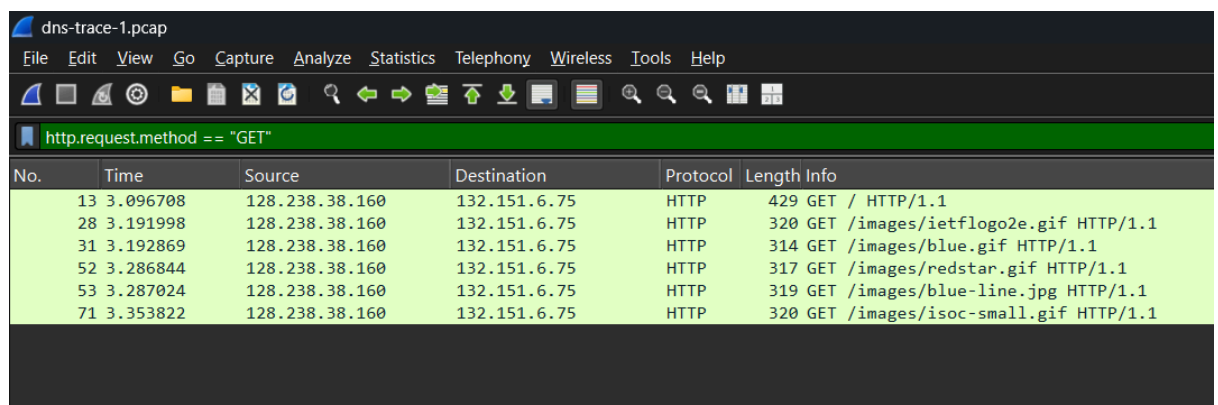
- Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Yes



- This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No

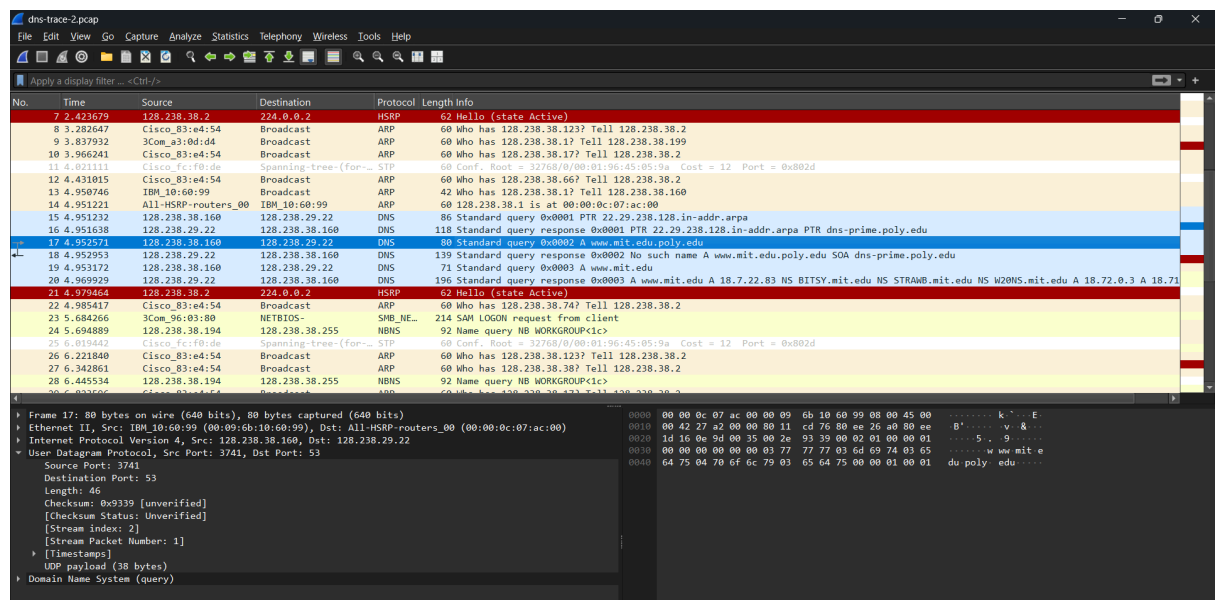
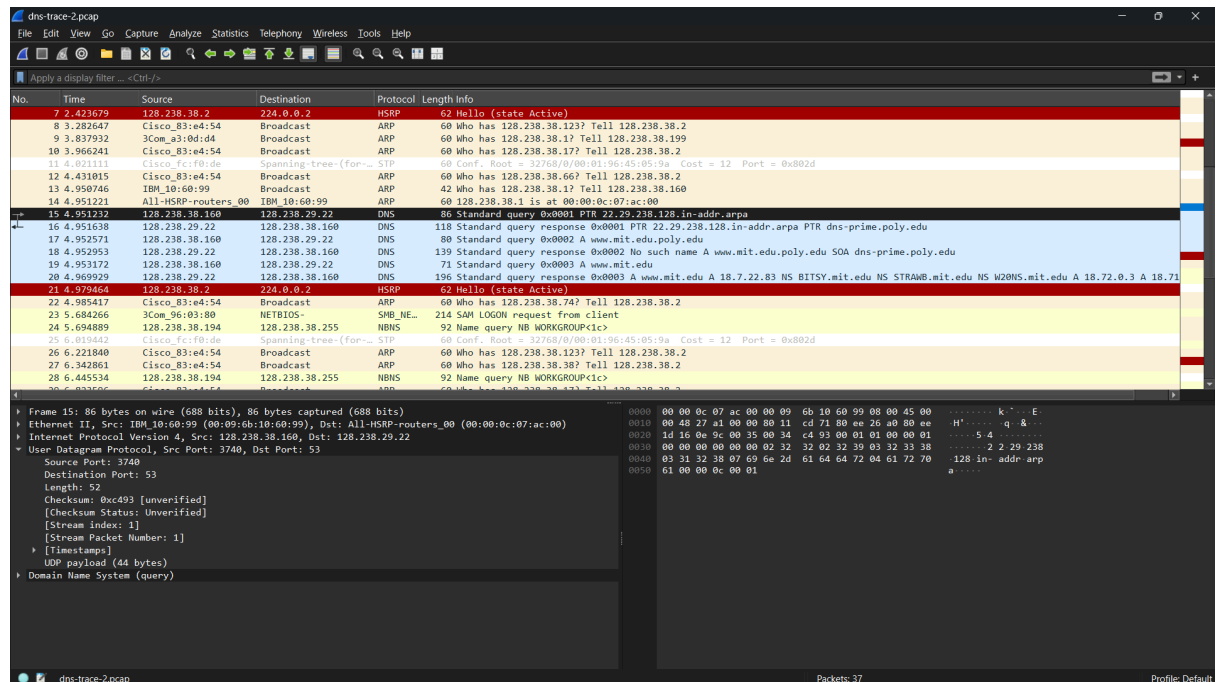


- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

DNS query destination port: 53

DNS response source port: 53



dns-trace-2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
7	2.423679	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
8	3.282647	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.123? Tell 128.238.38.2
9	3.837932	3Com:a3:0d:d4	Broadcast	ARP	60	Who has 128.238.38.17? Tell 128.238.38.199
10	3.966241	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.17? Tell 128.238.38.2
11	4.921111	Cisco:f:f0:de	Spanning-tree (for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
12	4.431015	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.66? Tell 128.238.38.2
13	4.950746	IBM:10:60:99	Broadcast	ARP	42	Who has 128.238.38.17? Tell 128.238.38.160
14	4.951221	All-HSRP-routers_00	IBM:10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu NS W20NS.mit.edu A 18.7.22.83
21	4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22	4.985417	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2
23	5.684266	3Com:96:03:80	NETBIOS-	SMB_NE...	214	SAM LOGON request from client
24	5.694889	128.238.38.194	128.238.38.255	NBNS	92	Name query NB WORKGROUP<1c>
25	6.019442	Cisco:f:f0:de	Spanning-tree (for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
26	6.221840	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.123? Tell 128.238.38.2
27	6.342861	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.38? Tell 128.238.38.2
28	6.445534	128.238.38.194	128.238.38.255	NBNS	92	Name query NB WORKGROUP<1c>

Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface 0

Ethernet II, Src: IBM:10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22

User Datagram Protocol, Src Port: 3742, Dst Port: 53

Source Port: 3742

Destination Port: 53

Length: 37

Checksum: 0x5890 [unverified]

[Checksum Status: Unverified]

[Stream Index: 3]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (29 bytes)

Domain Name System (query)

0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 08 00 45 00 k... T... E...
0010 00 39 27 a3 00 00 00 11 cd 7e 80 ee 26 a0 80 ee h... A...
0020 1d 16 0e 9e 00 35 00 25 58 00 03 01 00 00 01 5 % X...
0030 00 00 00 00 00 03 77 77 03 6d 69 74 03 65 www mit e...
0040 64 75 00 00 01 00 01 du...

dns-trace-2.pcap

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No.	Time	Source	Destination	Protocol	Length	Info
7	2.423679	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
8	3.282647	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.123? Tell 128.238.38.2
9	3.837932	3Com:a3:0d:d4	Broadcast	ARP	60	Who has 128.238.38.17? Tell 128.238.38.199
10	3.966241	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.17? Tell 128.238.38.2
11	4.921111	Cisco:f:f0:de	Spanning-tree (for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
12	4.431015	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.66? Tell 128.238.38.2
13	4.950746	IBM:10:60:99	Broadcast	ARP	42	Who has 128.238.38.17? Tell 128.238.38.160
14	4.951221	All-HSRP-routers_00	IBM:10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu NS W20NS.mit.edu A 18.7.22.83
21	4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22	4.985417	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2
23	5.684266	3Com:96:03:80	NETBIOS-	SMB_NE...	214	SAM LOGON request from client
24	5.694889	128.238.38.194	128.238.38.255	NBNS	92	Name query NB WORKGROUP<1c>
25	6.019442	Cisco:f:f0:de	Spanning-tree (for-...	STP	60	Conf. Root = 32768/0/00:01:96:45:05:9a Cost = 12 Port = 0x802d
26	6.221840	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.123? Tell 128.238.38.2
27	6.342861	Cisco:83:e4:54	Broadcast	ARP	60	Who has 128.238.38.38? Tell 128.238.38.2
28	6.445534	128.238.38.194	128.238.38.255	NBNS	92	Name query NB WORKGROUP<1c>

Frame 16: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0

Ethernet II, Src: Cisco:83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM:10:60:99 (00:09:6b:10:60:99)

Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160

User Datagram Protocol, Src Port: 53, Dst Port: 3740

Source Port: 53

Destination Port: 3740

Length: 84

Checksum: 0x2539 [unverified]

[Checksum Status: Unverified]

[Stream Index: 1]

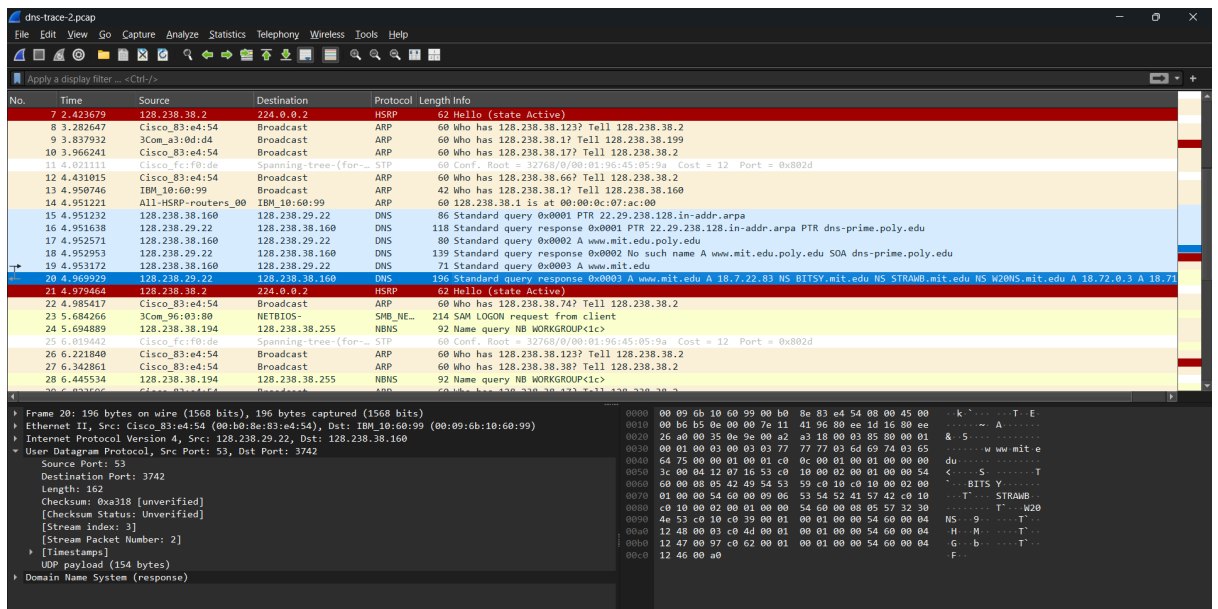
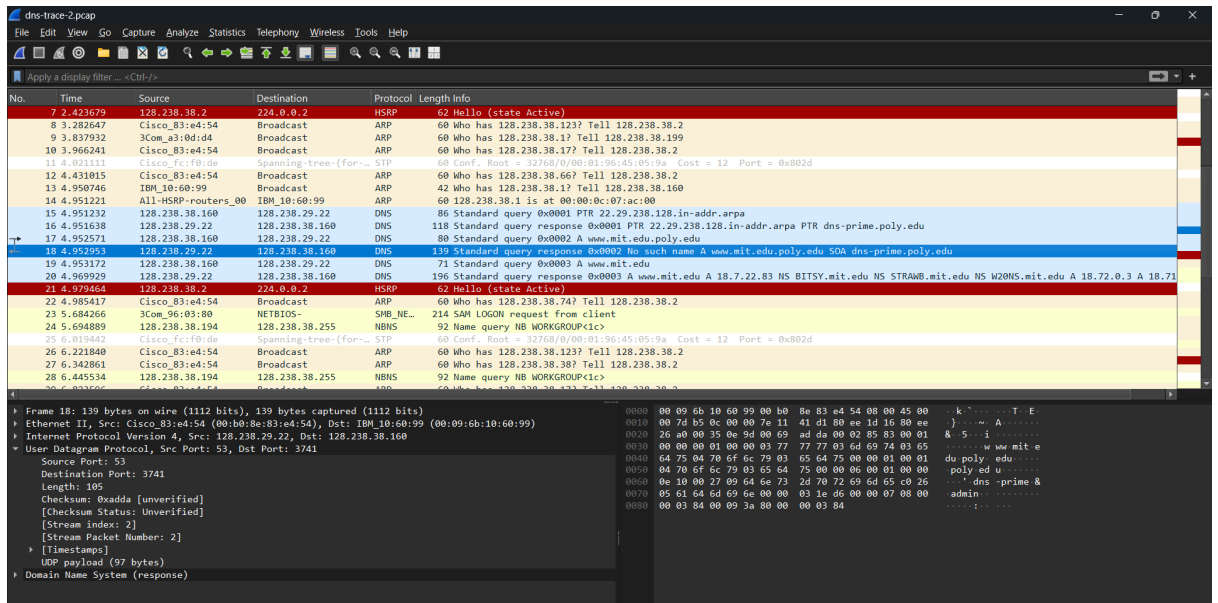
[Stream Packet Number: 2]

[Timestamps]

UDP payload (76 bytes)

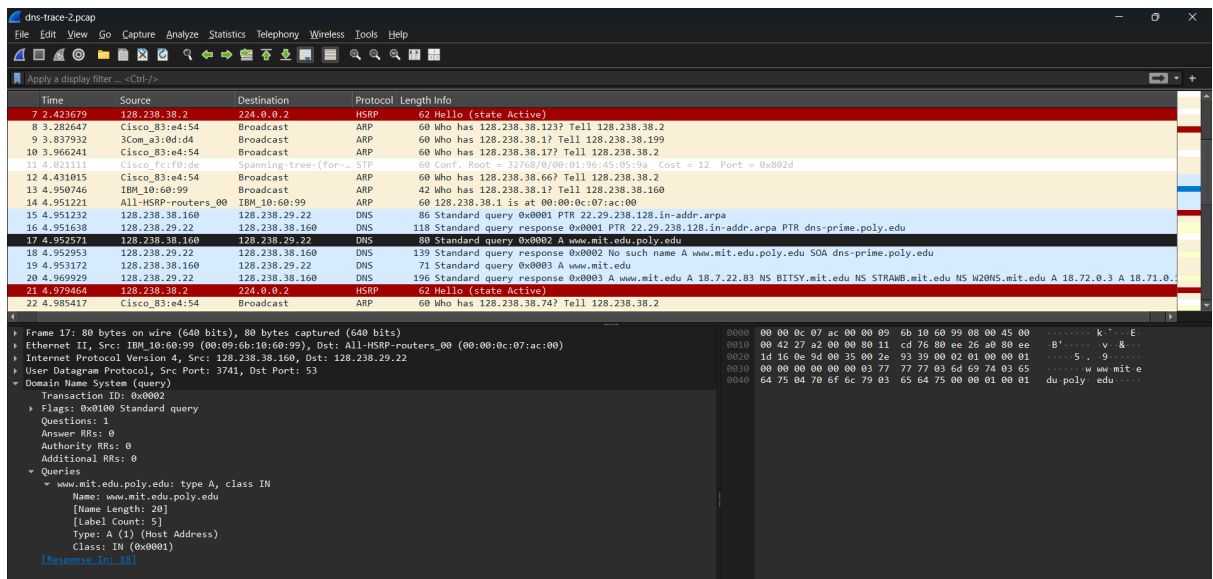
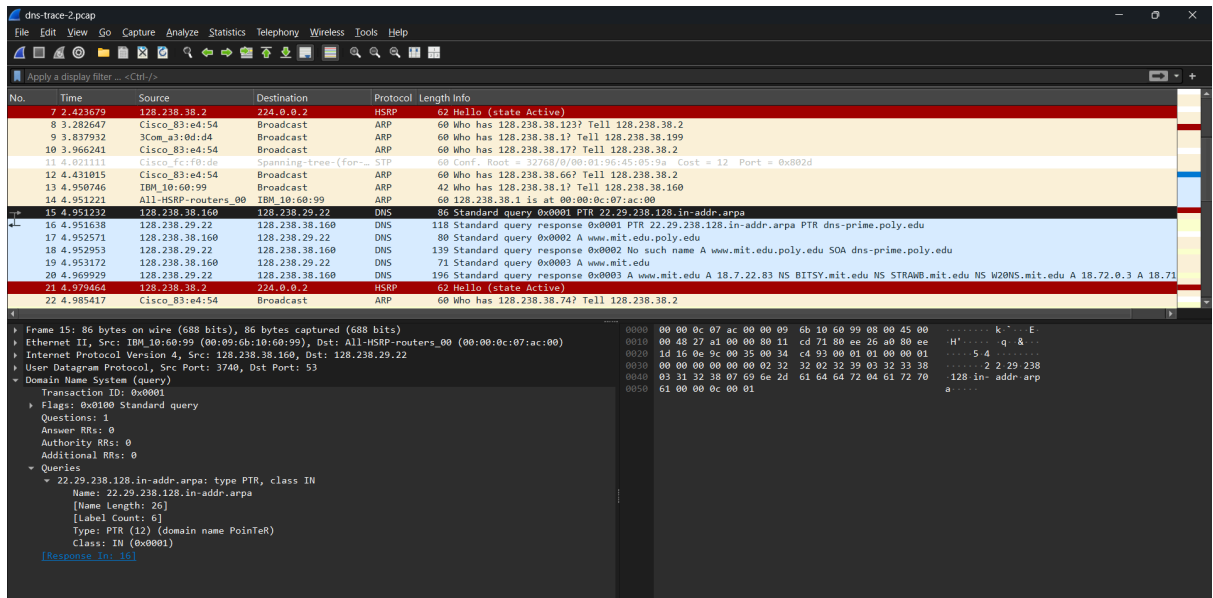
Domain Name System (response)

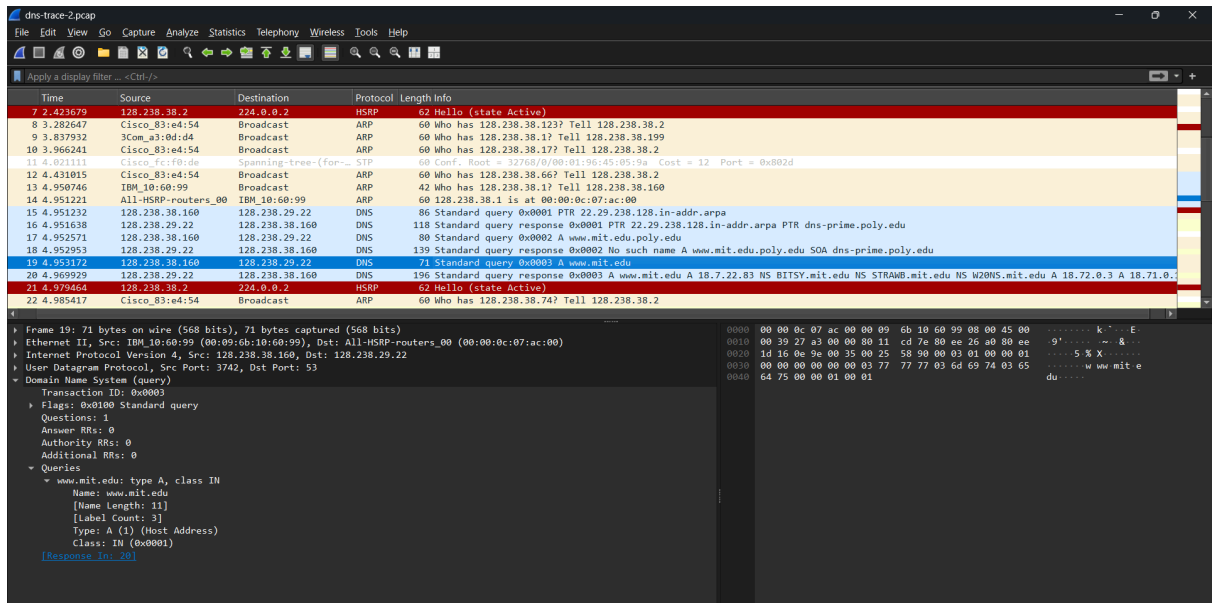
0000 00 09 6b 10 60 99 00 00 8e 83 e4 54 08 00 45 00 k... T... E...
0010 00 68 b5 0b 00 00 7e 11 41 e7 80 ee 1d 16 80 ee h... A...
0020 26 a0 00 35 0e 9c 00 54 25 39 00 01 85 80 00 01 & 5... T %9...
0030 00 01 00 00 00 00 02 32 32 02 32 39 01 32 33 38 2 2 29 238...
0040 03 31 32 38 07 69 0e 2d 61 64 64 72 04 61 72 70 128 In- addr arp
0050 61 00 00 0c 00 01 c0 0c 00 0c 00 01 00 00 0e 10 a...
0060 00 14 09 64 6e 73 2d 70 72 69 6d 65 04 70 6f 6c dns-p rime pol
0070 79 03 65 64 75 00 y edu



9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

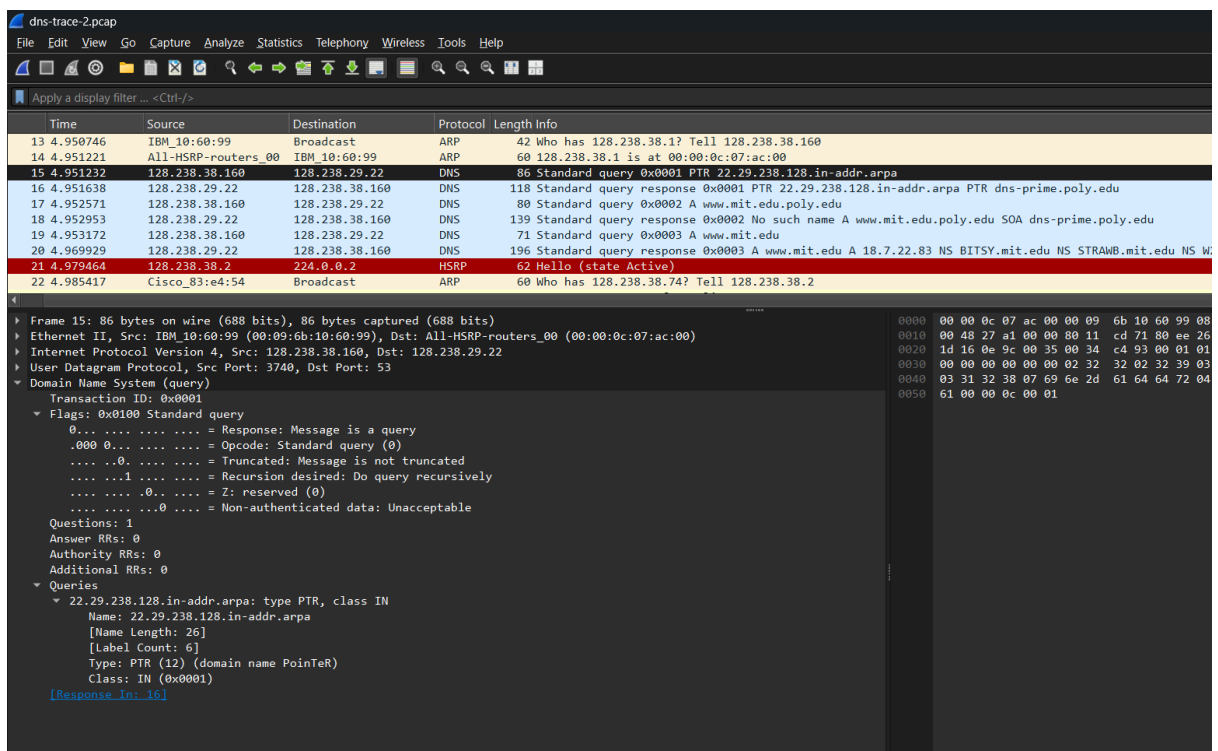
128.238.29.22, No





10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type PTR and A. No “answers”



dns-trace-2.pcap

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Time	Source	Destination	Protocol	Length	Info
13 4.950746	IBM_10:60:99	Broadcast	ARP	42	Who has 128.238.38.1? Tell 128.238.38.160
14 4.951221	All-HSRP-routers_00	IBM_10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15 4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16 4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17 4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18 4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19 4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20 4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRANB.mit.edu
21 4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22 4.985417	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2

Frame 17: 80 bytes on wire (640 bits), 80 bytes captured (640 bits)

Ethernet II, Src: IBM_10:60:99 (00:00:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22

User Datagram Protocol, Src Port: 3741, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x0002

Flags: 0x0100 Standard query

0... .. = Response: Message is a query

.000 0... .. = Opcode: Standard query (0)

.... .. = Truncated: Message is not truncated

.... ..1 = Recursion desired: Do query recursively

.... ..0... .. = Z: reserved (0)

.... ..0... .. = Non-authenticated data: Unacceptable

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

www.mit.edu.poly.edu: type A, class IN

Name: www.mit.edu.poly.edu

[Name Length: 20]

[Label Count: 5]

Type: A (1) (Host Address)

Class: IN (0x0001)

[Response in: 10]

dns-trace-2.pcap

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Apply a display filter ... <Ctrl-/>

Time	Source	Destination	Protocol	Length	Info
13 4.950746	IBM_10:60:99	Broadcast	ARP	42	Who has 128.238.38.1? Tell 128.238.38.160
14 4.951221	All-HSRP-routers_00	IBM_10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15 4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16 4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17 4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18 4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19 4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20 4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRANB.mi
21 4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22 4.985417	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2

Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)

Ethernet II, Src: IBM_10:60:99 (00:00:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22

User Datagram Protocol, Src Port: 3742, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x0003

Flags: 0x0100 Standard query

0... .. = Response: Message is a query

.000 0... .. = Opcode: Standard query (0)

.... .. = Truncated: Message is not truncated

.... ..1 = Recursion desired: Do query recursively

.... ..0... .. = Z: reserved (0)

.... ..0... .. = Non-authenticated data: Unacceptable

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

www.mit.edu: type A, class IN

Name: www.mit.edu

[Name Length: 11]

[Label Count: 3]

Type: A (1) (Host Address)

Class: IN (0x0001)

[Response in: 20]

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

First response: 1 “answers”

Second response: 0 “answers”

Third response: 1 “answers”

dns-trace-2.pcap

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Time	Source	Destination	Protocol	Length	Info
13.4.950746	IBM_10:60:99	Broadcast	ARP	42	Who has 128.238.38.1? Tell 128.238.38.160
14.4.951221	All-HSRP-routers_00	IBM_10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15.4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16.4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17.4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18.4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19.4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20.4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB
21.4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22.4.985417	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2

Frame 16: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0

Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)

Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160

User Datagram Protocol, Src Port: 53, Dst Port: 3740

Domain Name System (response)

Transaction ID: 0x0001

Flags: 0x8580 Standard query response, No error

- 1... .. = Response: Message is a response
- 0000... .. = Opcode: Standard query (0)
- ...1... .. = Authoritative: Server is an authority for domain
- ...0... .. = Truncated: Message is not truncated
- ...1... .. = Recursion desired: Do query recursively
- ...1... .. = Recursion available: Server can do recursive queries
- ...0... .. = Z: reserved (0)
- ...0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server
- ...0... .. = Non-authenticated data: Unacceptable
- ...0000... .. = Reply code: No error (0)

Questions: 1

Answer RRs: 1

Authority RRs: 0

Additional RRs: 0

Queries

- 22.29.238.128.in-addr.arpa: type PTR, class IN
 - Name: 22.29.238.128.in-addr.arpa
 - [Name Length: 26]
 - [Label Count: 6]
 - Type: PTR (12) (domain name Pointer)
 - Class: IN (0x0001)

Answers

- 22.29.238.128.in-addr.arpa: type PTR, class IN, dns-prime.poly.edu

[Request In: 15]

[Time: 0.000406000 seconds]

Number of answers in packet (dns.count.answers), 2 bytes

Packets: 37

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Time	Source	Destination	Protocol	Length	Info
13 4.950746	IBM_10:60:99	Broadcast	ARP	42	Who has 128.238.38.1? Tell 128.238.38.160
14 4.951221	All-HSRP-routers_00	IBM_10:60:99	ARP	60	128.238.38.1 is at 00:00:0c:07:ac:00
15 4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16 4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17 4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18 4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19 4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20 4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS 5
21 4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)
22 4.985417	Cisco_83:e4:54	Broadcast	ARP	60	Who has 128.238.38.74? Tell 128.238.38.2

Frame 18: 139 bytes on wire (1112 bits), 139 bytes captured (1112 bits)

Ethernet II, Src: Cisco 83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM 10:60:99 (00:09:6b:10:60:99)

Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160

User Datagram Protocol, Src Port: 53, Dst Port: 3741

Domain Name System (response)

Transaction ID: 0x0002

Flags: 0x8583 Standard query response, No such name

1... .. = Response: Message is a response

.000 0... .. = Opcode: Standard query (0)

... .1... .. = Authoritative: Server is an authority for domain

... ..0... .. = Truncated: Message is not truncated

... ..1... .. = Recursion desired: Do query recursively

... ..1... .. = Recursion available: Server can do recursive queries

... ..0... .. = Z: reserved (0)

... ..0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server

... ..0... .. = Non-authenticated data: Unacceptable

... ..0011 = Reply code: No such name (3)

Questions: 1

Answer RRs: 0

Authority RRs: 1

Additional RRs: 0

Queries

www.mit.edu.poly.edu: type A, class IN

Name: www.mit.edu.poly.edu

[Name Length: 20]

[Label Count: 5]

Type: A (1) (Host Address)

Class: IN (0x0001)

Authoritative nameservers

poly.edu: type SOA, class IN, mname dns-prime.poly.edu

[Request In: 17]

[Time: 0.000382000 seconds]

Number of answers in packet (dns.count.answers), 2 bytes

Packets: 37

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Time	Source	Destination	Protocol	Length	Info
19 4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20 4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu
21 4.979464	128.238.38.2	224.0.0.2	HSRP	62	Hello (state Active)

Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)

Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)

Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160

User Datagram Protocol, Src Port: 53, Dst Port: 3742

Domain Name System (response)

Transaction ID: 0x0003

Flags: 0x8580 Standard query response, No error

1... .. = Response: Message is a response

.000 0... .. = Opcode: Standard query (0)

....1... .. = Authoritative: Server is an authority for domain

....0... .. = Truncated: Message is not truncated

....1... .. = Recursion desired: Do query recursively

....1... .. = Recursion available: Server can do recursive queries

....0... .. = Z: reserved (0)

....0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server

....0... .. = Non-authenticated data: Unacceptable

....0000 = Reply code: No error (0)

Questions: 1

Answer RRs: 1

Authority RRs: 3

Additional RRs: 3

Queries

www.mit.edu: type A, class IN

Name: www.mit.edu

[Name Length: 11]

[Label Count: 3]

Type: A (1) (Host Address)

Class: IN (0x0001)

Answers

www.mit.edu: type A, class IN, addr 18.7.22.83

Authoritative nameservers

mit.edu: type NS, class IN, ns BITSY.mit.edu

mit.edu: type NS, class IN, ns STRAWB.mit.edu

mit.edu: type NS, class IN, ns W20NS.mit.edu

Additional records

BITSY.mit.edu: type A, class IN, addr 18.72.0.3

STRAWB.mit.edu: type A, class IN, addr 18.71.0.151

W20NS.mit.edu: type A, class IN, addr 18.70.0.160

[Request in: 19]

[Time: 0.016757000 seconds]

Number of answers in packet (dns.count.answers), 2 bytes

Packets: 37